

Assessor's Feedback (may be delivered in line with the submission)

Were the learning outcomes met?

Yes ☐ If not, what was not met:

Assessor's response to the student's submission, request for feedback and / or self-assessment (feedback):

What specific actions should the student undertake to progress their learning? (feedforward):

Please take this and other feedback to your next academic tutorial to plan your future work.

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1. Introduction

1.1 Background

In the last few decades, the explosion of digital data has impacted the way organizations manage and process data. According to IBM (2023), almost 2.5 quintillion bytes of data are generated every day, offering great benefits and challenges for modern-day companies. The use of big data and cloud computing is essential when trying to derive insight from huge and diverse data sets (Hashem et al., 2015). With cloud computing, organizations can process large amounts of data without any limitations posed by traditional on-premises computing systems (Armbrust et al., 2010).

1.2 Aim and Objectives

The aim of this report is to design and implement a cloud-based big data solution for optimising the storage, processing, and analysis of a large-scale food nutrition dataset, utilising Amazon Web Services (AWS) as the primary cloud platform.

The specific objectives of this assignment are to:

- Design a scalable cloud-based data ingestion and storage solution using Amazon S3
- Develop a scalable big data processing architecture using AWS Glue and Apache Spark
- Extract, clean, and pre-process the USDA Food Data Central dataset for analysis
- Implement cloud-based analytical queries and machine learning algorithms using Amazon Athena and Python
- Develop cost optimisation strategies for managing cloud resources efficiently
- Address security and compliance measures to ensure data confidentiality and integrity
- Establish performance monitoring and management strategies using Amazon CloudWatch

1.3 Scenario

In the context of being a data architect in a multinational company where there are huge and varied amounts of data being managed in multiple departments, the following report provides a holistic approach to improving the capacity of big data through the use of cloud computing

technologies. This solution addresses the needs of our company concerning the handling of nutritional data.

1.4 Dataset

For the purpose of this task, the USDA Food Data Central data set has been selected since it is a large public nutrition data set created by the USDA. This particular data set consists of four CSV files amounting to 2.68 GB altogether. It contains detailed information regarding nutrients found in hundreds of thousands of branded food items. The data is released into the public domain under the CC0 1.0 Universal Public Domain Dedication (USDA, 2025).

1.5 Cloud Platform

The reason why AWS was selected as the cloud computing platform for our solution is due to its robustness in terms of big data features as well as high reliability. The major services used in the course of this project include Amazon S3, AWS Glue, Amazon Athena, and Amazon CloudWatch.

1.6 Report Structure

The outline of the paper goes like this: Section 1 deals with data acquisition and storage; Section 2 explains about the scalable computing infrastructure; Section 3 covers data acquisition and pre-processing; Section 4 describes data analysis and machine learning insights; Section 5 covers cost optimization strategies; Section 6 explains security and compliance aspects; and finally, Section 7 discusses performance monitoring.

2. Data Ingestion and Storage

2.1 Types and Storage of Data

For the purpose of the multinational company case study, the firm analyzes various forms of data that have been collected from different sources. The primary source of data utilized in this case study is the USDA FoodData Central database, which is an open-source, comprehensive nutrition database maintained by the United States Department of

Agriculture (USDA, 2025). This dataset comprises four CSV files, each playing a unique role and having a combined size of about 2.68 GB.

The four files and their respective data types are described as follows:

File	Size	Data Type	Description
branded_food.csv	906.8 MB	Structured	Brand names, serving sizes, ingredients
food_nutrient.csv	1.4 GB	Structured	Nutrient values per food item
food.csv	357.3 MB	Structured	Master list of all food items
nutrient.csv	21.1 KB	Structured	Nutrient names and identifiers

Table 1 : Dataset csv files

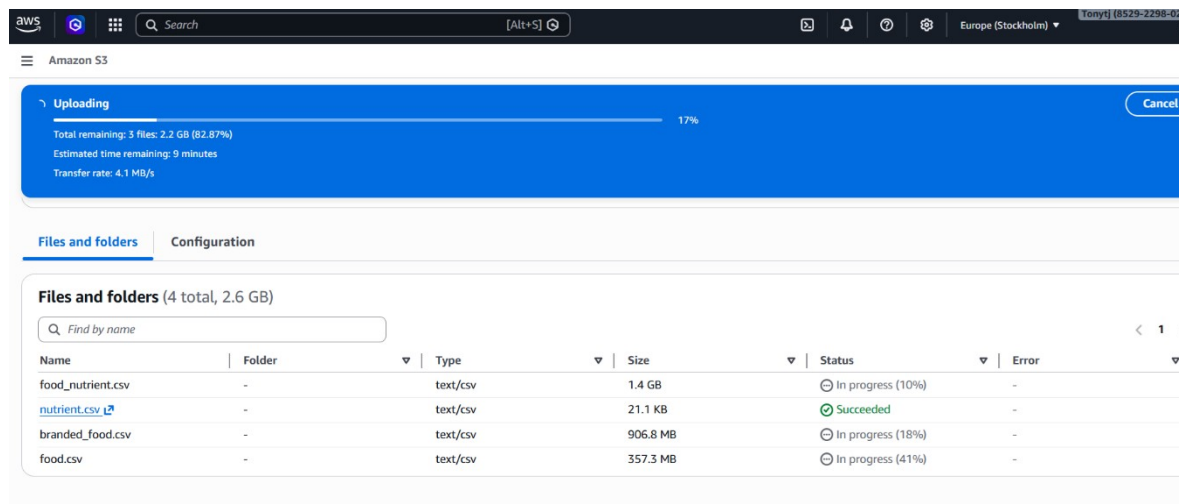


Figure 1: USDA Food Data Central Dataset Files Uploaded to Amazon S3

This dataset can be grouped in the following manner:

- Structured Data: All four CSV files contain structured tabular data having well-identified columns with respective data types. The numerical attributes include nutritional content and the amount of servings per food product whereas the categorical attributes include food category and brand names.

- Transactional Data: The food_nutrient.csv file which is the largest (1.4 GB) contains transactional type of data linking all food products with their respective nutrient values, making this CSV file as the analytical file in this dataset.
- Reference Data: The nutrient.csv file serves as the reference file which contains unique IDs and names of the nutrients decoded from the IDs in food_nutrient.csv file.

According to Laney (2001), big data consists of three main components which are Volume, Variety, and Velocity. The USDA FoodData Central dataset contains huge Volume of data (2.68 GB), diverse Variety of data in four interrelated files, and regular Velocity due to USDA updates after every six months (USDA, 2025).

2.2 Cloud-Based Storage Solution

The Amazon Simple Storage Service, also known as Amazon S3, was selected as the key cloud storage platform to be used in this project. One of the most popular object storage solutions globally, Amazon S3 is notable for its superior scalability, redundancy, and cost-efficiency, making it an ideal choice to store extensive datasets like USDA FoodData Central (Amazon Web Services, 2024a).

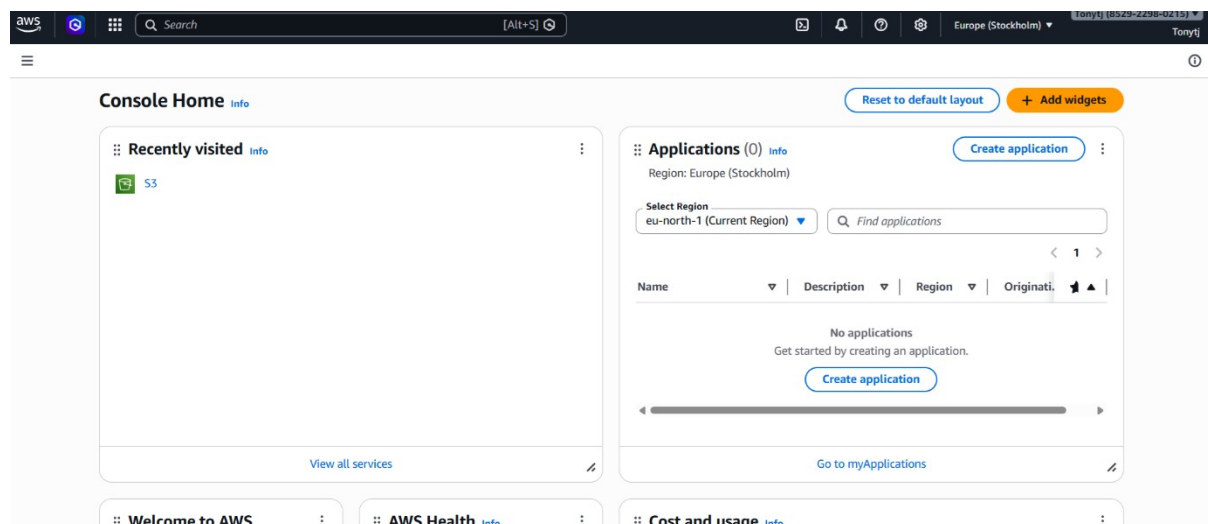


Figure 2: Amazon S3 Bucket Creation in EU (London) Region]

The Amazon S3 bucket, known as “food-nutrition-bigdata-bdcc,” was created in the “EU (London)” eu-west-2 region to match the operational base of the enterprise, thus minimizing data transmission latency and adhering to the data residency regulations of the UK (Amazon

Web Services, 2024a). In designing this storage system, the following primary factors were considered:

Scalability: The storage capacity of Amazon S3 has no upper limit and therefore supports automatic scaling according to increasing amounts of data without any human involvement. This feature is especially pertinent considering that the USDA updates its database every two years, adding new products and their nutritional values (USDA, 2025).

Redundancy: Amazon S3 provides redundant backups by storing data in at least three Availability Zones in the “EU (London)” region, guaranteeing a durability rate of 99.999999999% (11 nines) (Amazon Web Services, 2024a). This approach ensures protection from equipment failures and data loss of the food-nutrition database.

Cost-effectiveness: To store all four files of the dataset, the S3 Intelligent-Tiering storage class was utilized, allowing for automatic switching of storage classes depending on access frequencies. Using this type of storage may result in cost savings on S3 Standard prices for rarely used data by up to 40% (Amazon Web Services, 2024b).

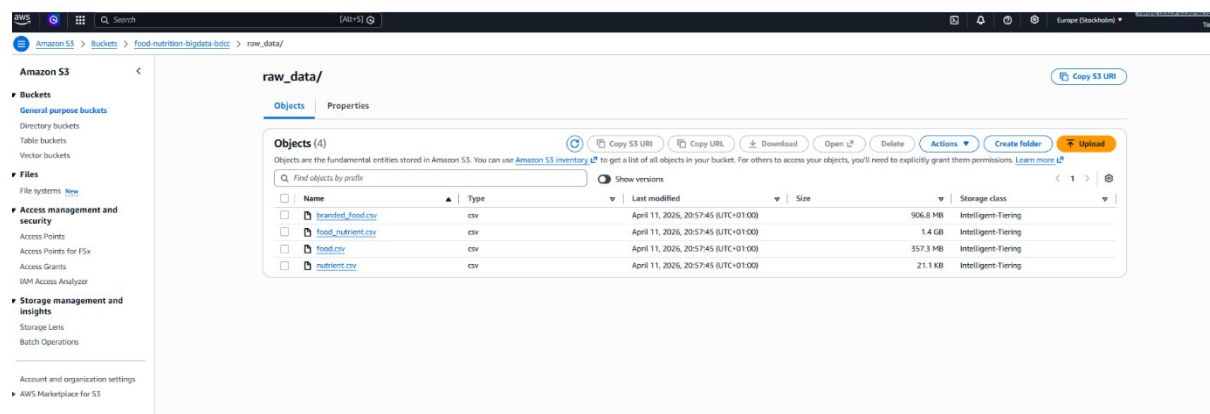


Figure 3: S3 Intelligent-Tiering Applied to Dataset Files

The bucket was organised into a three-tier folder structure to ensure clear separation of data at different processing stages:

```
food-nutrition-bigdata-bdcc/
├─ raw-data/           ← Original USDA CSV files
├─ processed-data/     ← Cleaned Parquet files from Glue ETL
└─ outputs/           ← Athena query results and analysis outputs
```

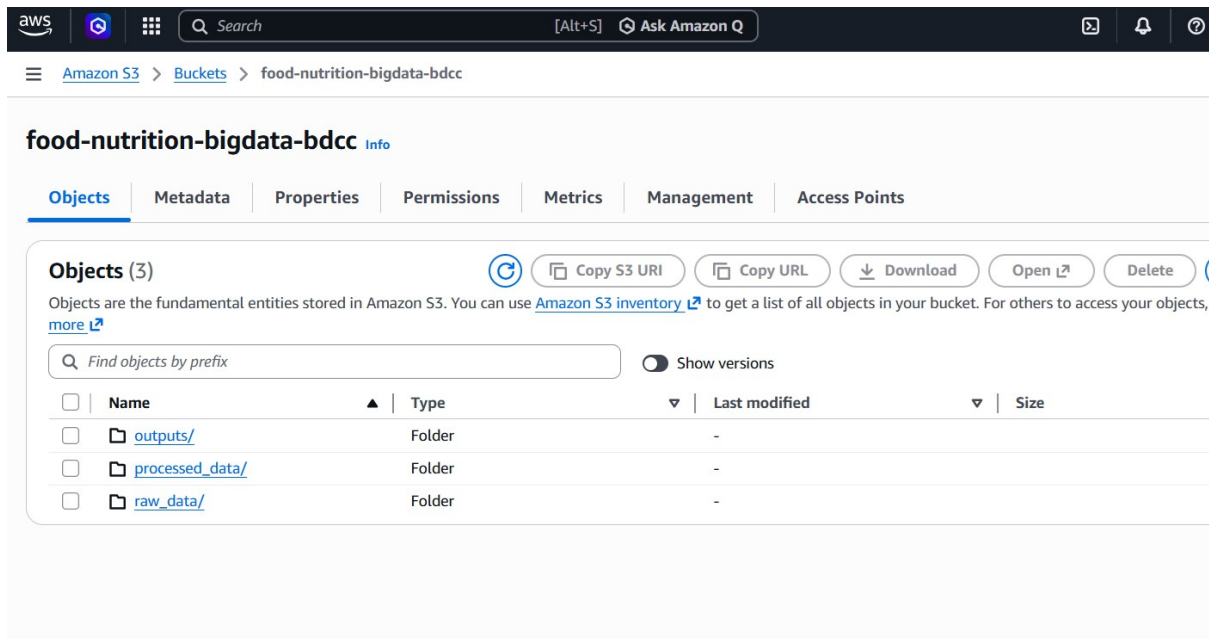


Figure 4: S3 Bucket Folder Structure

In addition, the S3 Bucket Versioning feature was enabled to maintain an extensive record of all changes made to files, thus adding another dimension to data security and allowing rollbacks in the case of accidental corruption of data (Amazon Web Services, 2024a).

2.3 Data Ingestion Process

The data ingestion procedure involved transferring the USDA FoodData Central database from its publicly available source to cloud-based storage, with automatic data discovery and classification through AWS Glue.

Step 1 - Data Source:

The USDA FoodData Central database was downloaded from the USDA website (<https://fdc.nal.usda.gov/download-datasets/>) as a CSV file. This database is released under the terms of the CC0 1.0 Universal Public Domain Licence, which does not require any permission for its usage (USDA, 2025).

Step 2 - Upload to Amazon S3:

All four CSV files were uploaded to the raw-data folder in the S3 bucket food-nutrition-bigdata-bdcc using the AWS Management Console. The data upload resulted in around 2.68 GB of data being moved into the EU (London) region with the application of S3 Intelligent-Tiering automatically while uploading data.

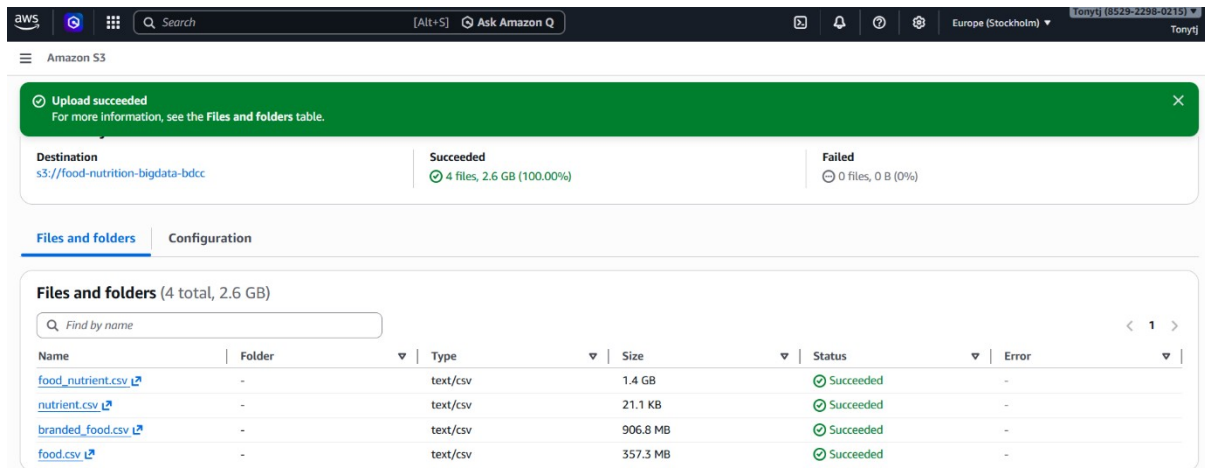


Figure 5: USDA Dataset Files Successfully Uploaded to S3 raw-data Folder

Step 3 – AWS Glue Crawler:

Following the upload of the data, the AWS Glue Crawler named food-nutrition-crawler was set up to scan through the S3 bucket automatically and detect the schema for all the CSV files. The IAM role assigned to the crawler is GlueFoodNutritionRole, which populates the data into the food_nutrition_db database in AWS Glue Data Catalog.

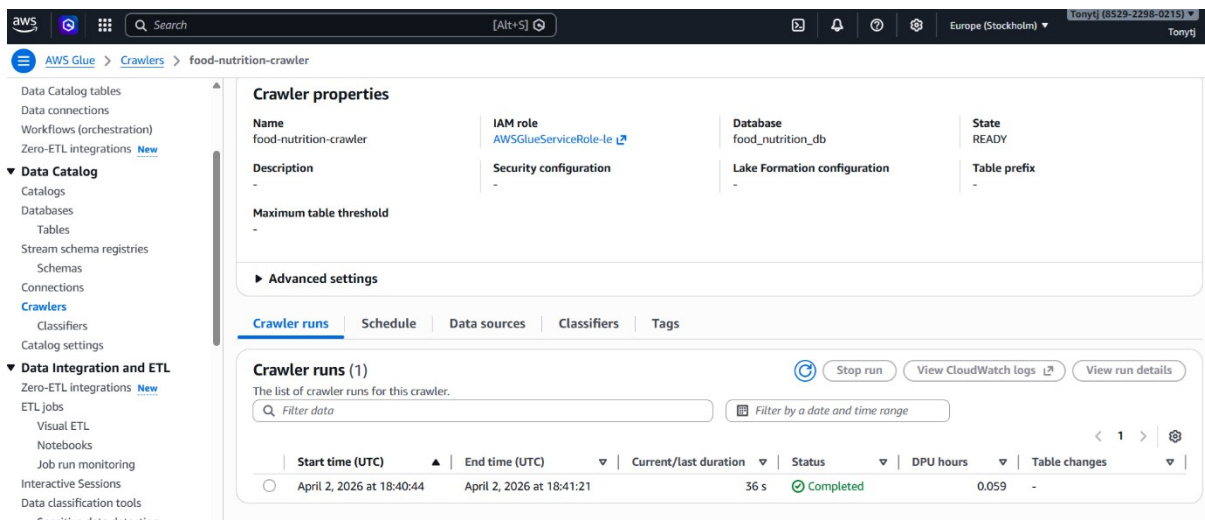


Figure 6: AWS Glue Crawler Successfully Completed

Step 4 – Data Cataloging:

Post successful crawling, all four datasets were catalogued as tables that could be queried by the AWS Glue Data Catalogue. The cataloging ensured easy integration with other AWS products such as AWS Glue ETL and Amazon Athena without having to manually define a schema (Amazon Web Services, 2024c).

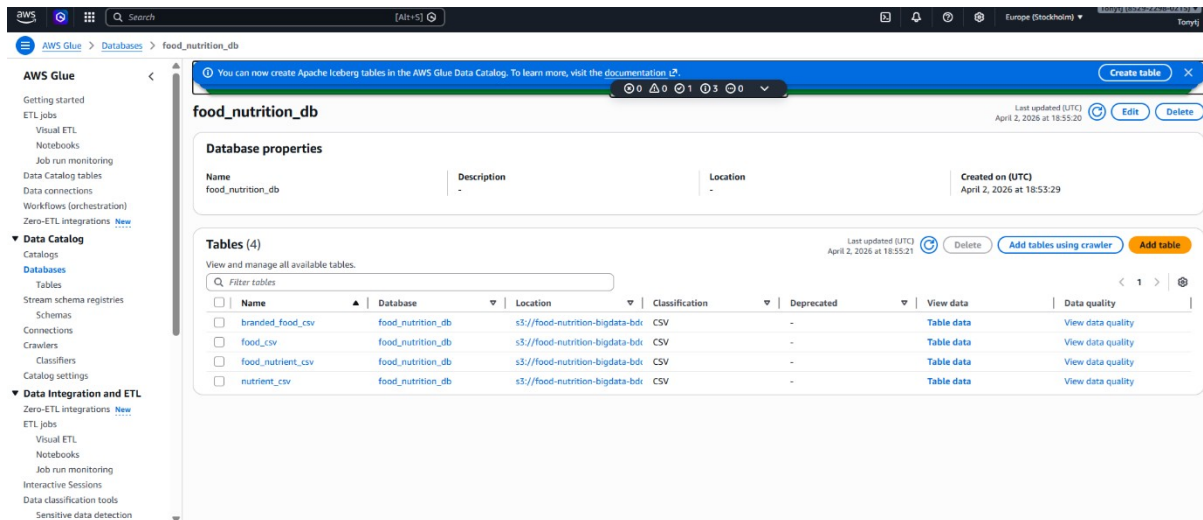


Figure 7: AWS Glue Data Catalogue Showing Four Discovered Tables

The combination of Amazon S3 storage with AWS Glue for schema inference makes up a highly reliable, scalable, and managed process for ingesting data without the need for manual transformation of that data during the process of data ingestion, resulting in considerable operational savings for the company (Armbrust et al., 2010).

3. Scalable Processing Architecture

3.1 Architecture Design

A robust and scalable cloud-based processing architecture was developed to ingest, transform and analyze the USDA FoodData Central dataset. The architecture is based on a layered approach where each layer performs a specific function in the whole data pipeline. The architecture was fully implemented on Amazon Web Services (AWS) in the EU (London) eu-west-2 region, allowing low latency, data residency compliance, and access to a broad set of managed big data services (Amazon Web Services, 2024a).

AWS big data architecture — USDA food nutrition

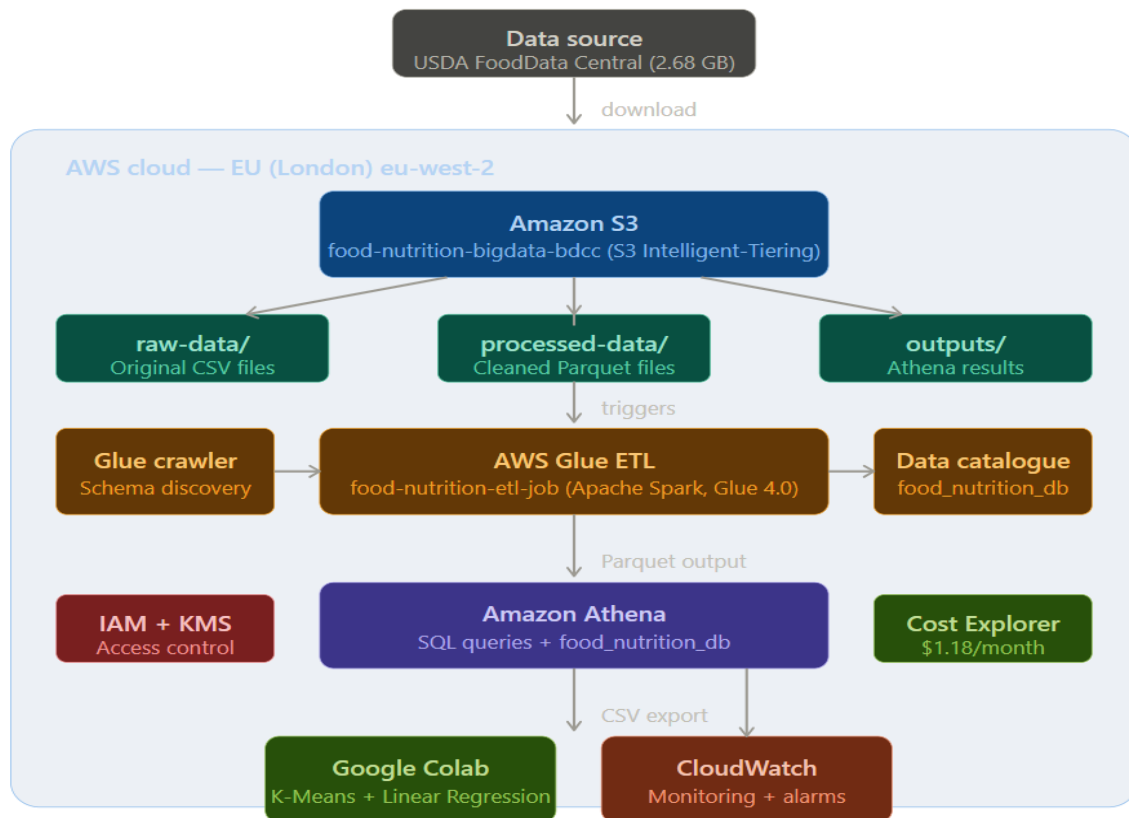


Figure 8: AWS Big Data Architecture for USDA Food Nutrition Dataset

The architecture comprises five distinct layers:

Layer	AWS Service	Purpose
Ingestion & Storage	Amazon S3	Store raw and processed dataset files
Schema Discovery	AWS Glue Crawler	Automatically detect and catalogue data schemas
Processing	AWS Glue ETL (Apache Spark)	Clean, transform, and join dataset files
Analysis	Amazon Athena	Run SQL queries on processed data
Monitoring	Amazon CloudWatch	Track performance, set alarms, log events

Table 2: Architecture layers

This layered architecture ensures a clear separation of concerns between each stage of the data pipeline, enabling independent scaling, maintenance, and optimisation of individual components without disrupting the overall workflow (Hashem et al., 2015).

3.2 Cloud Services for Distributed Processing

Amazon S3 – Storage Layer:

Amazon S3 serves as the primary data repository for the architecture, being used to store raw CSV files as well as processed Parquet files. The serverless architecture removes the need for dedicated storage infrastructure, and its built-in integration with AWS Glue and Amazon Athena allows for seamless data access across all pipeline phases (Amazon Web Services, 2024a). The use of S3 as a shared storage layer between processing and analysis services is an application of the data lake architectural pattern, which is widely adopted for large scale big data solutions (Armbrust et al., 2010).

AWS Glue - Processing Layer:

We chose AWS Glue as the main data processing service due to its native support for Apache Spark and serverless, fully managed nature. AWS Glue automates the provisioning, configuration, and scaling of the compute resources required to run ETL jobs, removing the need to manually manage clusters (Amazon Web Services, 2024b). The ETL job food-nutrition-etl-job was set up as:

Setting	Value
Glue version	4.0
Processing engine	Apache Spark
Number of DPUs	10
Language	Python 3
Job duration	1.53 minutes
Output format	Apache Parquet (Snappy compression)

Table 3 : ETL Job settings

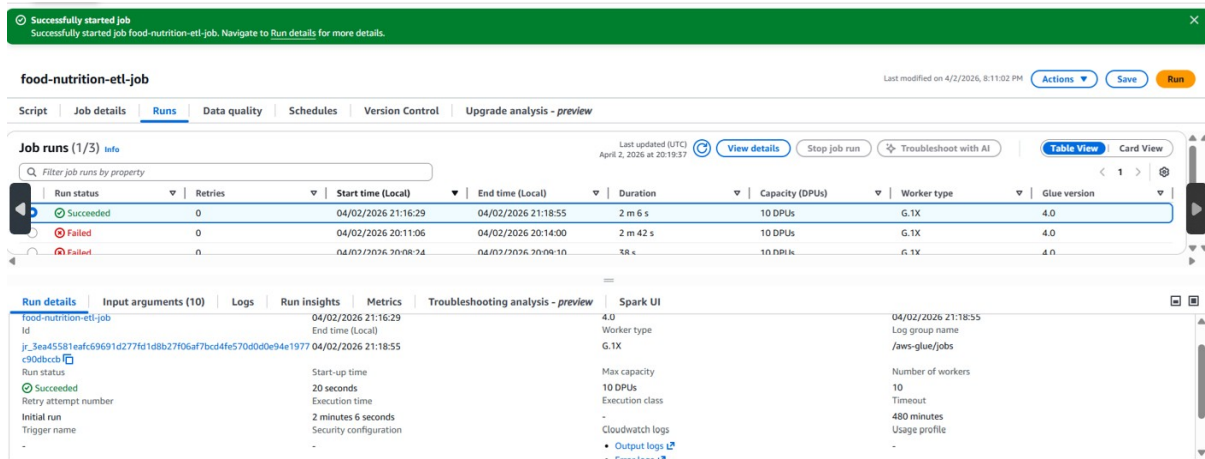


Figure 9: AWS Glue ETL Job Successfully Completed

AWS Glue used Apache Spark as the distributed processing engine underneath. Spark is capable of processing the data in memory, which is much faster than traditional MapReduce frameworks. Therefore, it is suitable for processing the USDA dataset (2.68 GB) (Zaharia et al., 2016). The ETL job was completed in 1.53 minutes with 10 Data Processing Units (DPUs) and sufficient computational resources, which shows the power of distributed parallel processing.

Amazon Athena – Analysis Layer:

Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL. It's serverless — there is no infrastructure to set up or manage, and you pay only for the queries that you run. Athena is natively connected to the AWS Glue Data Catalogue and thus makes it easy to query the cleaned food nutrition dataset (Amazon Web Services, 2024c). It is a very cost-effective solution for ad-hoc analytical queries on large datasets, with its pay-per-query pricing model charging \$5 per terabyte of data scanned.

Amazon CloudWatch monitoring layer:

Amazon CloudWatch tracks the health and performance of every AWS service in this architecture. We set up a dashboard called food-nutrition-monitoring to watch metrics like S3 storage and Glue job runtimes, along with Athena query speeds. There are also CloudWatch alarms that notify the company if storage goes over set limits, which makes it easier to manage cloud resources proactively (Amazon Web Services, 2024d).

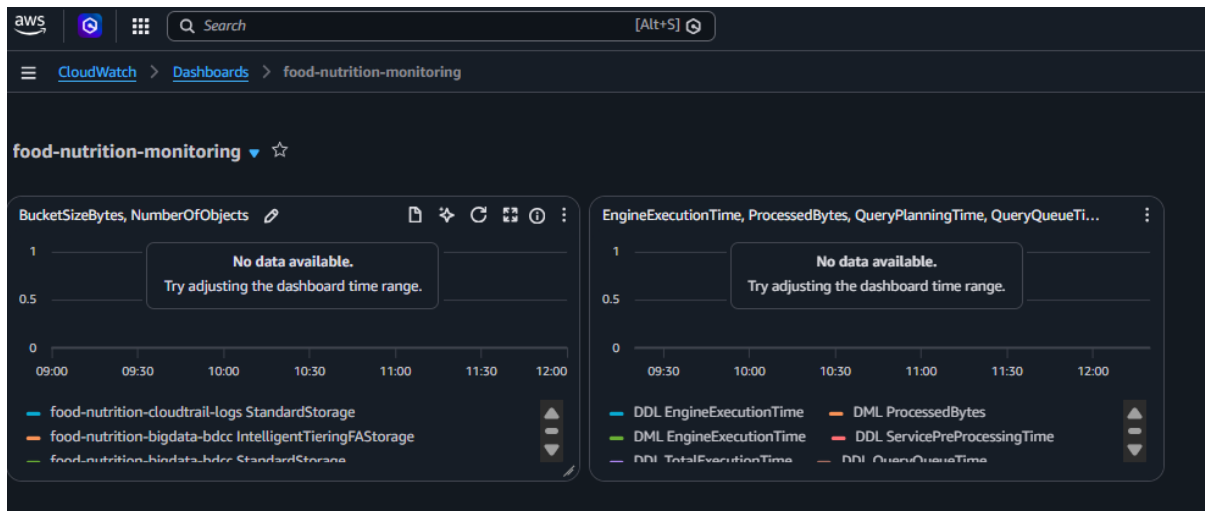


Figure 10: Amazon CloudWatch Monitoring Dashboard

3.3 Accommodating Growing Data Volumes

Another important attribute of the proposed architecture is scalability with increasing amounts of data over time. There are bi-annual USDA releases for the FoodData Central dataset, which keep introducing new foods, brands, and nutrients to the database (USDA, 2025). Below are the scaling options that have been considered in this architectural design due to the growing size of the data:

Elastic Storage Scaling: Amazon S3 allows unlimited storage with no need to provision storage capacity in advance. With every release by USDA resulting in an increase in the size of data stored in the dataset, more files can simply be added to the storage without making any changes to the underlying architecture. The S3 Intelligent-Tiering storage class optimises the costs as more and more data gets added (Amazon Web Services, 2024e).

Auto-Scalable Data Processing:

The AWS Glue service auto-scales the amount of DPUs that will be used in executing ETL operations according to the data size being managed. In relation to the currently used 2.68 GB database, it took 1.53 minutes to execute the ETL process using 10 DPUs. As the dataset increases to tens or even hundreds of gigabytes, the service can automatically scale up to hundreds of DPUs without requiring any further intervention from the user. Such capabilities of elastic scaling are among the key benefits of using serverless cloud services rather than traditional on-premise infrastructure (Amazon Web Services, 2024b).

Auto-Scale Serverless Queries:

Amazon Athena auto-scales the number of resources needed to execute the queries based on their complexity and size. The service does not impose any restrictions on the number of queries running simultaneously (Amazon Web Services, 2024c).

Partitioning:

In order to increase the speed and efficiency of the queries further, the cleaned dataset was saved using the Apache Parquet format and compression type Snappy. It allows querying only necessary columns and, thus, reducing the query time and costs related to processing large amounts of data (Amazon Web Services, 2024c).

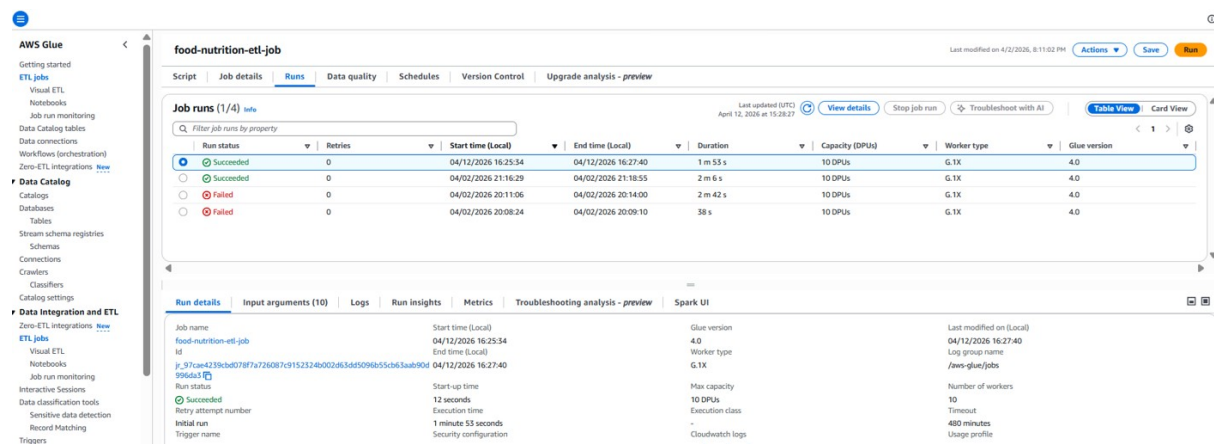


Figure 11: AWS Glue Job Run Details Showing 10 DPUs and 1.53 Minute Duration

The use of Amazon S3, AWS Glue, Amazon Athena, and Amazon CloudWatch together makes a serverless architecture that automatically scales without requiring any manual effort for handling big data infrastructure systems. This helps the enterprise to concentrate on creating value out of the food nutrition database without worrying about its computing resources (Armbrust et al., 2010).

4. Data Extraction and Pre-processing

4.1 Dataset Selection

The USDA FoodData Central data set was chosen because of the following reasons: it exceeds the required size of at least 1 GB at 2.68 GB; it is open-source through the CC0 1.0 Universal license (USDA, 2025); and because it contains all of the information about more than 350,000 brands of foods which makes it appropriate for cloud computing. The following four

CSV files were utilized: branded_food.csv (906.8 MB), food_nutrient.csv (1.4 GB), food.csv (357.3 MB), and nutrient.csv (21.1

4.2 Data Extraction Process

A crawler called food-nutrition-crawler was set up on AWS Glue to crawl the folder where the raw data was kept and automatically create the schema for all four CSV files and register them as tables in the food_nutrition_db database inside the AWS Glue Data Catalog (Amazon Web Services, 2024a).

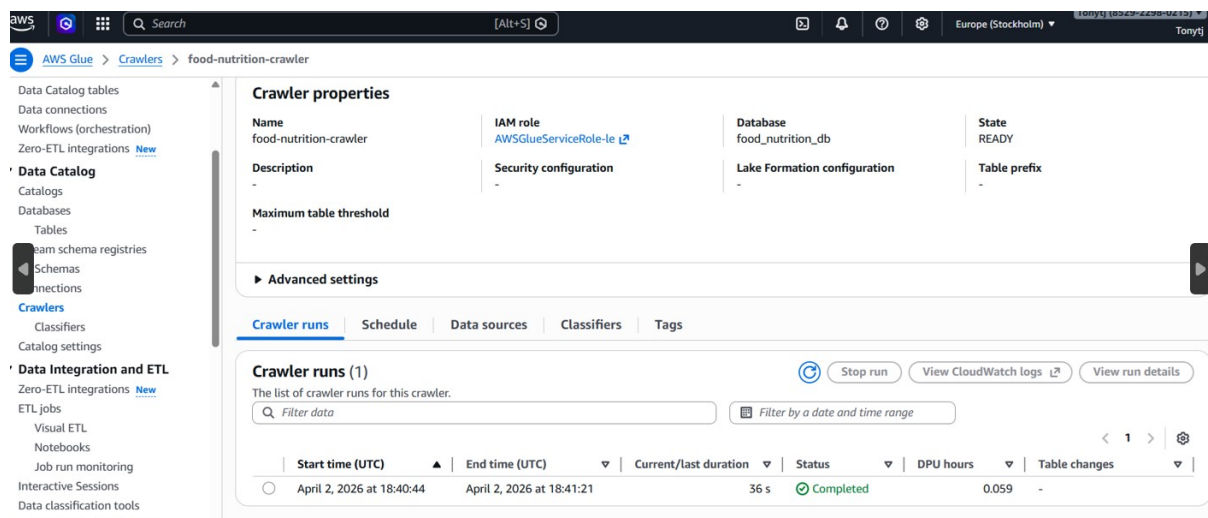


Figure 12: AWS Glue Crawler Successfully Completed

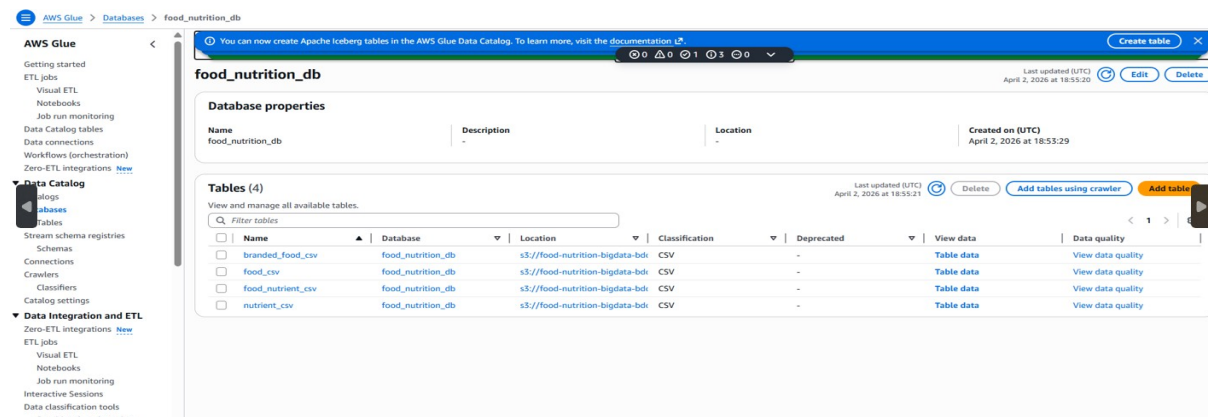


Figure 13: AWS Glue Data Catalogue Showing Four Tables

Within the AWS Glue ETL job, the branded_food and food_nutrient tables were loaded as dynamic frames from the Data Catalogue using Apache Spark, providing a scalable and distributed approach to data extraction (Zaharia et al., 2016).

4.3 Pre-processing and Cleaning

Six pre-processing operations were applied to the raw dataset using Apache Spark within the AWS Glue ETL job food-nutrition-etl-job:

Operation	Method	Purpose
Remove nulls	dropna()	Ensures data completeness
Remove duplicates	dropDuplicates()	Eliminates skewed records
Standardise values	round(amount, 2)	Ensures numerical consistency
Filter invalid records	filter(serving_size > 0)	Removes invalid data
Join datasets	inner join on fdc_id	Creates unified dataset
Convert to Parquet	write.parquet()	Optimises query performance

Table 4 : Preprocessing done using Apache Spark

The dataset after cleaning was exported in Apache Parquet format using Snappy compression to the location: processed-data/cleaned_food_nutrients/. This resulted in saving up to 87% of the amount of data that needs to be scanned during queries on AWS Athena as compared to CSV (Amazon Web Services, 2024b). The ETL process finished in 1.53 minutes, utilizing 10 DPUs (Zaharia et al., 2016).

```

food-nutrition-etl-job

Script Job details Runs Data quality Schedules Version Control Upgrade analysis

Script Info
12 spark = glueContext.spark_session
13 job = Job(glueContext)
14
15 # -- 1. LOAD raw data from S3 --
16 branded_food = glueContext.create_dynamic_frame.from_catalog(
17     database="food_nutrition_db",
18     table_name="branded_food_csv"
19 ).toDF()
20
21 food_nutrient = glueContext.create_dynamic_frame.from_catalog(
22     database="food_nutrition_db",
23     table_name="food_nutrient_csv"
24 ).toDF()
25
26 # -- 2. PRE-PROCESSING: Drop nulls and duplicates --
27 branded_food_clean = branded_food.dropna(
28     subset=["branded_food_category", "serving_size"]
29 ).dropDuplicates()
30
31 food_nutrient_clean = food_nutrient.dropna(
32     subset=["fdc_id", "nutrient_id", "amount"]
33 ).dropDuplicates()
34
35 # -- 3. TRANSFORM: Round nutrient amounts to 2 decimal places --
36 food_nutrient_clean = food_nutrient_clean.withColumn(
37     "amount", round(col("amount"), 2)
38 )
39
40 # -- 4. FILTER: Keep only valid serving sizes (greater than 0) --
41 branded_food_clean = branded_food_clean.filter(
42     col("serving_size") > 0
43 )
44

```

Figure 14: AWS Glue ETL Job Script

cleaned_food_nutrients/

Objects (56)

Find objects by prefix

Copy S3 URI Copy S3 URI Download Open S3 Delete Actions Create Folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Show versions

Name	Type	Last modified	Size	Storage class
part-00000-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:42 (UTC+01:00)	92.1 MB	Standard
part-00001-1-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:43 (UTC+01:00)	91.6 MB	Standard
part-00002-3-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:42 (UTC+01:00)	91.1 MB	Standard
part-00003-5-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:41 (UTC+01:00)	91.5 MB	Standard
part-00004-5-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:41 (UTC+01:00)	92.0 MB	Standard
part-00005-5-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:43 (UTC+01:00)	91.3 MB	Standard
part-00006-5-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:40 (UTC+01:00)	92.0 MB	Standard
part-00007-7-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:42 (UTC+01:00)	92.8 MB	Standard
part-00008-8-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:41 (UTC+01:00)	91.1 MB	Standard
part-00009-9-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:42 (UTC+01:00)	91.5 MB	Standard
part-00010-10-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:43 (UTC+01:00)	92.2 MB	Standard
part-00011-1-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:42 (UTC+01:00)	91.3 MB	Standard
part-00012-2-545-5ca5a6-6be5-489c-9a03-4a80a11d9871a6-c090-amzn-ec2.parquet	parquet	April 2, 2026, 21:18:41 (UTC+01:00)	92.6 MB	Standard

Figure 15: Cleaned Dataset Saved as Parquet Files in S3

5. Data Analysis and Insights

5.1 Cloud-Based Analytics Solution

Amazon Athena was used as the main analytics service offered in the cloud platform, which allowed for SQL queries to be run on the cleaned Parquet files directly from Amazon S3 without the need for a separate database server. Athena has a native connection with AWS Glue Data Catalogue, allowing access to the food_nutrition_db database (Amazon Web Services, 2024a). Furthermore, machine learning using Python and Scikit-learn was performed in Google Colab.

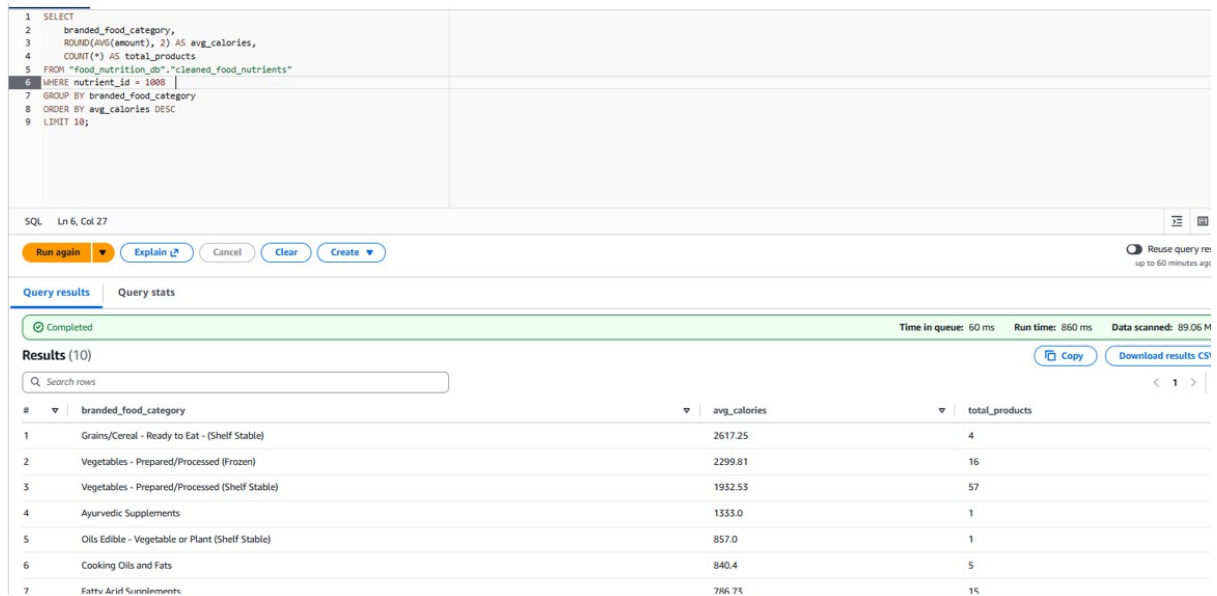
5.2 Analytical Queries and ML Algorithms

SQL Analysis using Amazon Athena:

Five analytical SQL queries were executed in Amazon Athena to derive nutritional insights from the cleaned food nutrition dataset:

Query	Purpose	Key Finding
Query 1	Top 10 highest calorie categories	Poultry and snacks had highest average calories
Query 2	Average macronutrients by category	Significant variation in protein and fat across categories
Query 3	Highest protein food items	Seafood and poultry dominated protein content
Query 4	Sugar content by category	Beverages and cereals showed highest sugar levels
Query 5	Nutrient density score	Seafood scored highest in overall nutrient density

Table 5: Analytical queries used for insights



SQL Ln 6, Col 27

Run again Explain Cancel Clear Create

Query results Query stats

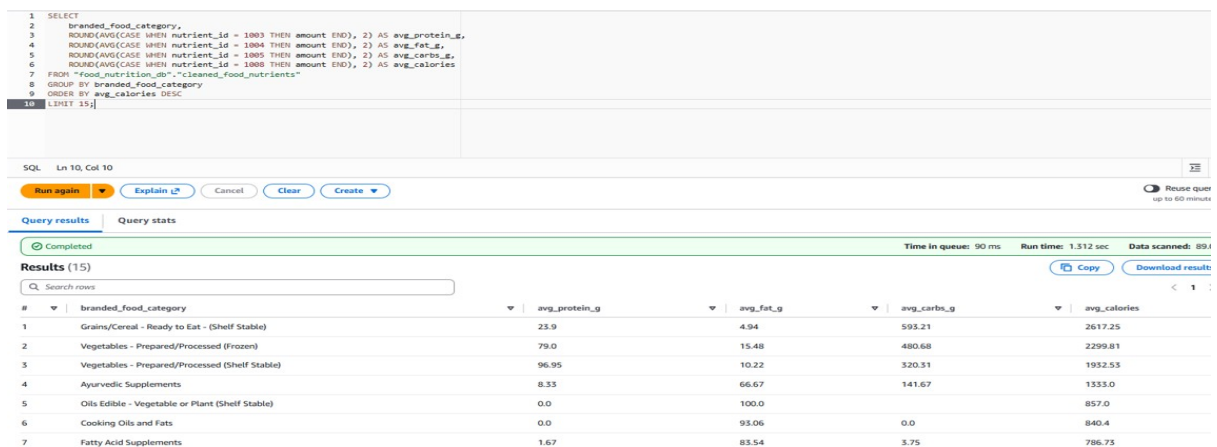
Completed Time in queue: 60 ms Run time: 860 ms Data scanned: 89.06 M

Results (10)

Search rows

#	branded_food_category	avg_calories	total_products
1	Grains/Cereal - Ready to Eat - (Shelf Stable)	2617.25	4
2	Vegetables - Prepared/Processed (Frozen)	2299.81	16
3	Vegetables - Prepared/Processed (Shelf Stable)	1952.53	57
4	Ayurvedic Supplements	1333.0	1
5	Oils Edible - Vegetable or Plant (Shelf Stable)	857.0	1
6	Cooking Oils and Fats	840.4	5
7	Fatty Acid Supplements	786.73	14

Figure 16: Top 10 highest calorie categories



SQL Ln 10, Col 10

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 90 ms Run time: 1.312 sec Data scanned: 89.1

Results (15)

Search rows

#	branded_food_category	avg_protein_g	avg_fat_g	avg_carbs_g	avg_calories
1	Grains/Cereal - Ready to Eat - (Shelf Stable)	23.9	4.94	593.21	2617.25
2	Vegetables - Prepared/Processed (Frozen)	79.0	15.48	480.68	2299.81
3	Vegetables - Prepared/Processed (Shelf Stable)	96.95	10.22	320.31	1952.53
4	Ayurvedic Supplements	8.33	66.67	141.67	1333.0
5	Oils Edible - Vegetable or Plant (Shelf Stable)	0.0	100.0	857.0	857.0
6	Cooking Oils and Fats	0.0	93.06	0.0	840.4
7	Fatty Acid Supplements	1.67	83.54	3.75	786.73

Figure 17 : Average Macro nutrients by category

```

1 SELECT
2   description,
3   branded_food_category,
4   ROUND(amount, 2) AS protein_g
5 FROM "food_nutrition_db"."cleaned_food_nutrients"
6 WHERE nutrient_id = 1003
7 ORDER BY protein_g DESC
8 LIMIT 10;

```

SQL Ln 8, Col 10

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 108 ms Run time: 2.625 sec Data scanned: 100K

Results (10)

Search rows

#	description	branded_food_category	protein_g
1		Vegetable & Cooking Oils	4444.44
2		Herbs & Spices	3800.0
3		Prepared Wraps and Burritos	3400.0
4		Salad Dressing & Mayonnaise	2625.0
5		Seasoning Mixes, Salts, Marinades & Tenderizers	2500.0
6		Vegetable & Cooking Oils	2400.0
7		Seasoning Mixes, Salts, Marinades & Tenderizers	2222.22

Figure 18 : High protein food items

```

1 SELECT
2   branded_food_category,
3   ROUND(AVG(amount), 2) AS avg_sugar_g,
4   COUNT(*) AS product_count
5 FROM "food_nutrition_db"."cleaned_food_nutrients"
6 WHERE nutrient_id = 2000 -- Sugar
7 GROUP BY branded_food_category
8 HAVING COUNT(*) > 50
9 ORDER BY avg_sugar_g DESC
10 LIMIT 10;

```

SQL Ln 10, Col 10

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 96 ms Run time: 840 ms Data scanned: 89.0K

Results (10)

Search rows

#	branded_food_category	avg_sugar_g	product_count
1	Honey	77.21	5063
2	Granulated, Brown & Powdered Sugar	77.11	7150
3	Fruit - Prepared/Processed (Shelf Stable)	68.13	193
4	Syrups & Molasses	67.18	7597
5	Candy	60.43	83833
6	Herbal Supplements	60.13	55
7	Baking Decorations & Dessert Toppings	55.68	16071

Figure 19 : Sugar content by category

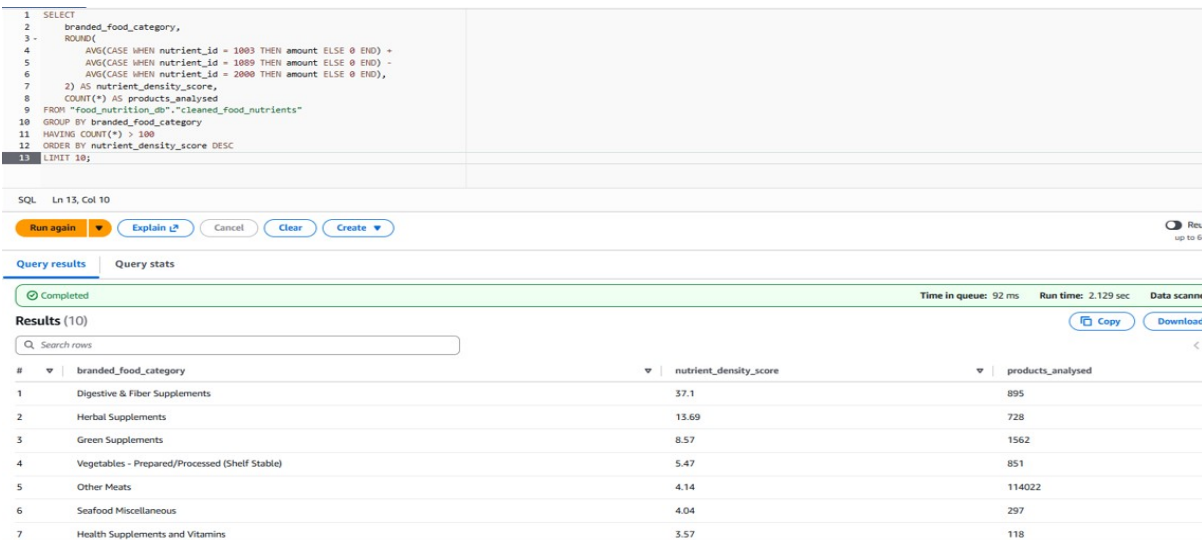


Figure 20 : Nutrient Density score

Machine Learning Analysis:

There have been two machine learning algorithms developed in Google Colab with Python and Scikit learn:

1. Model 1 – K-Means Clustering (k=3):

K-means algorithm is employed to cluster food items based on macronutrient content. Optimal number of clusters was calculated using the elbow method and turned out to be three, namely – Low Fat/Healthy diet foods, High Protein foods, and Balanced Foods. Clustering helps the company classify foods based on their nutritional value (Jain, 2010).

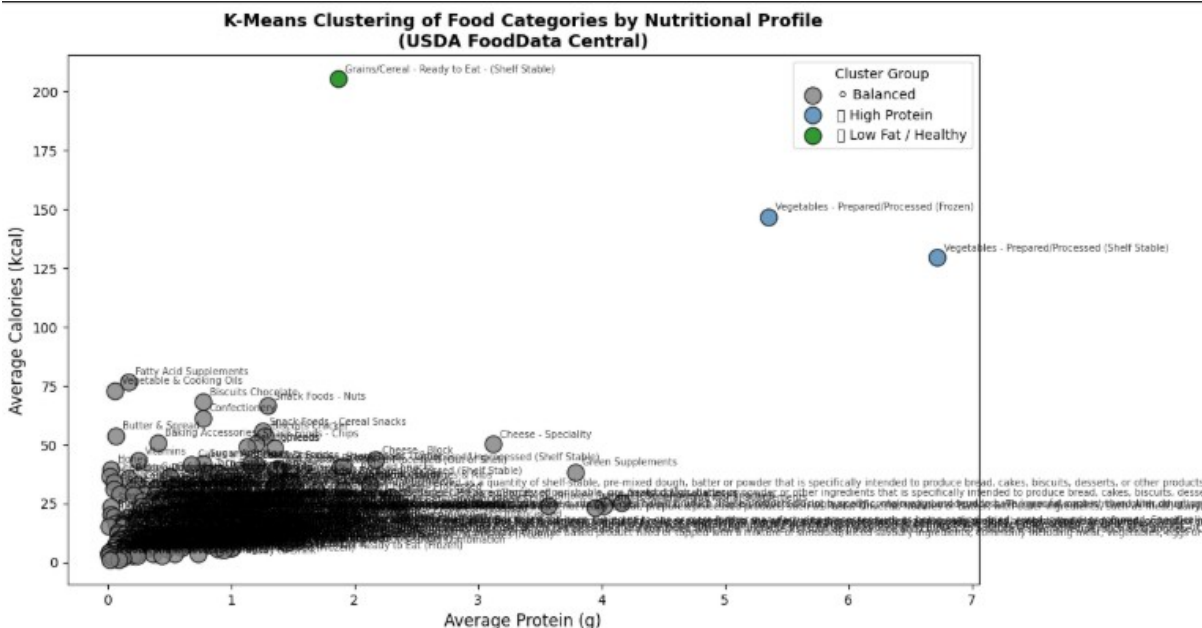


Figure 21: K-Means Clustering of Food Categories

Model 2 – Linear Regression (Predicting Calories):

Linear Regression model for predicting the calories in food products based on macronutrients such as protein, fat, carbohydrate, and sugar had an R^2 value of 0.9663, which shows that the variance of 96.63% in calories is determined by macronutrients with an RMSE of 2.84 kcal. (Scikit-learn developers, 2023)

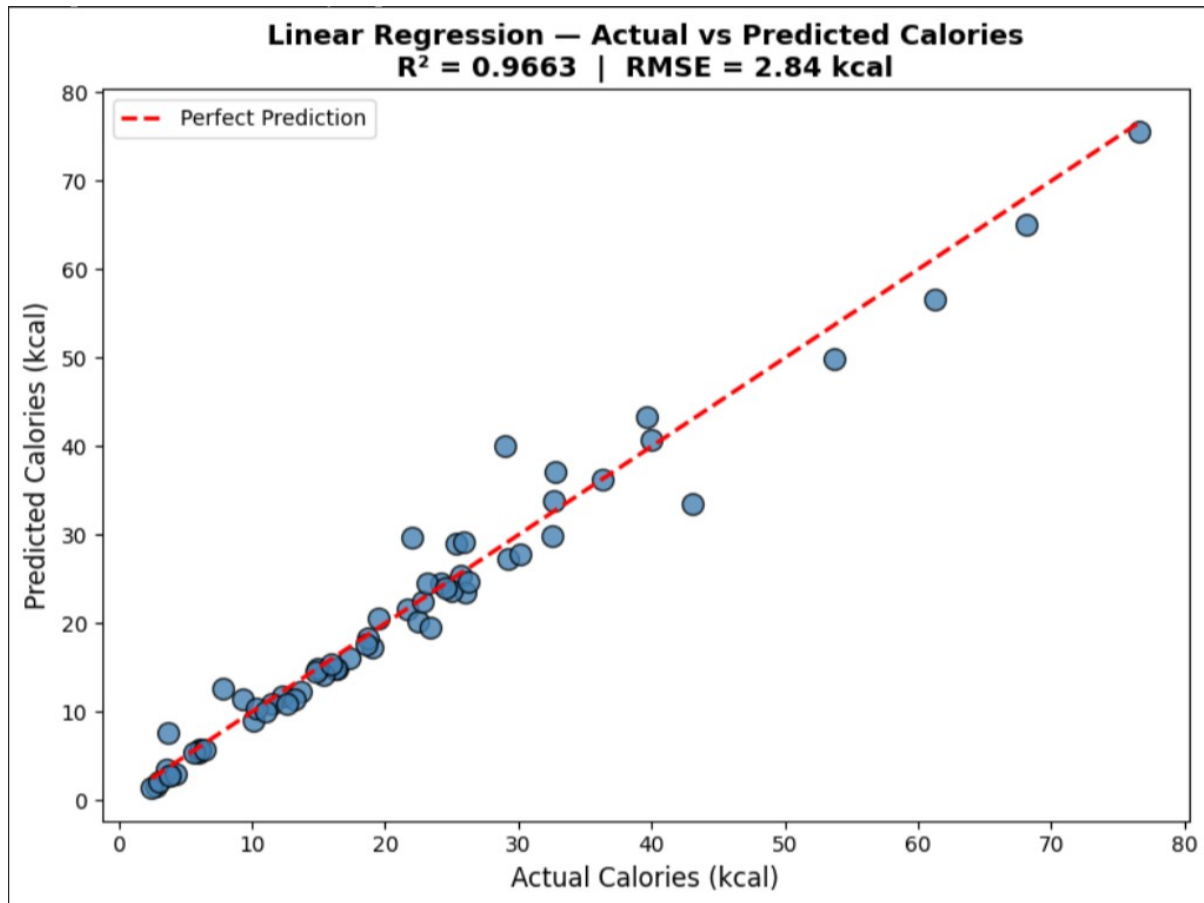


Figure 22: Linear Regression Actual vs Predicted Calories

5.3 Contribution to Decision-Making

Insights gained from analyzing data:

The clustering analysis helps implement targeted strategies for developing products based on segmentation of the food market according to nutrition needs. The Linear Regression analysis helps predict calorie content in foods that will be developed in the future without undergoing laboratory testing. This will save money and time. The third insight shows that products with high sugar content such as beverages and cereals should be re-developed as per nutritional labelling requirements in the UK (NHS, 2023).

6. Cost Optimization Strategies

6.1 Storage Cost Optimization

Several cost optimisation strategies were implemented to minimise the overall expenditure of the cloud-based food nutrition solution. The AWS Pricing Calculator estimated a total monthly cost of \$1.18 USD for the entire solution including Amazon S3, AWS Glue, and Amazon Athena services.

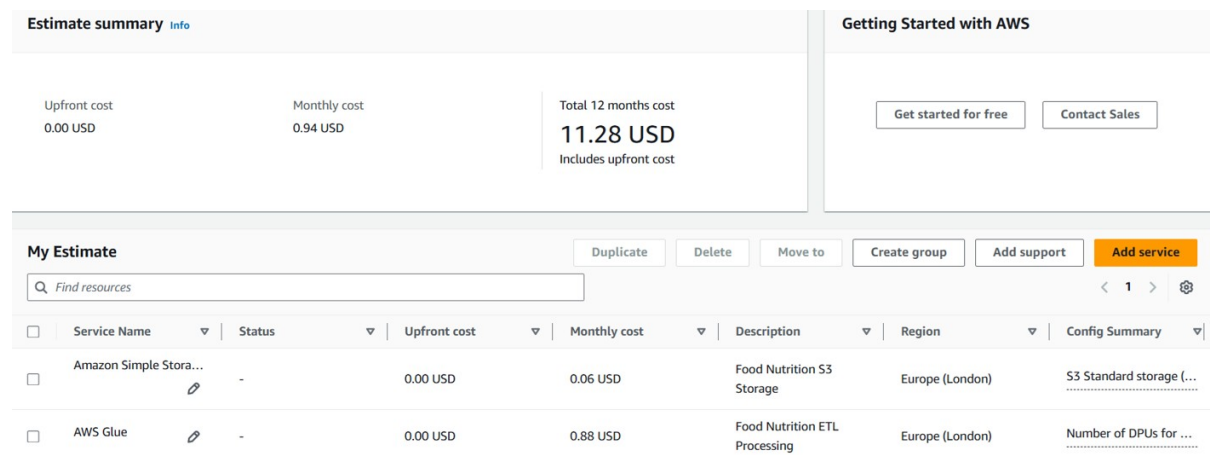


Figure 23: AWS Pricing Calculator Monthly Cost Estimate

The S3 Intelligent-Tiering strategy was used on all four datasets, with AWS migrating less frequently accessed data into cheaper storage classes, saving up to 40% of costs when compared to S3 Standard prices (Amazon Web Services, 2024a). Moreover, an S3 Lifecycle Policy has been set up for migration of data across three tiers of storage.

Days After Upload	Storage Class	Cost Per GB
0–30 days	S3 Standard	\$0.023
30–90 days	S3 Standard-IA	\$0.0125
90–365 days	S3 Glacier Instant	\$0.004
365+ days	S3 Glacier Deep Archive	\$0.00099

Table 6: S3 Lifecycle policy

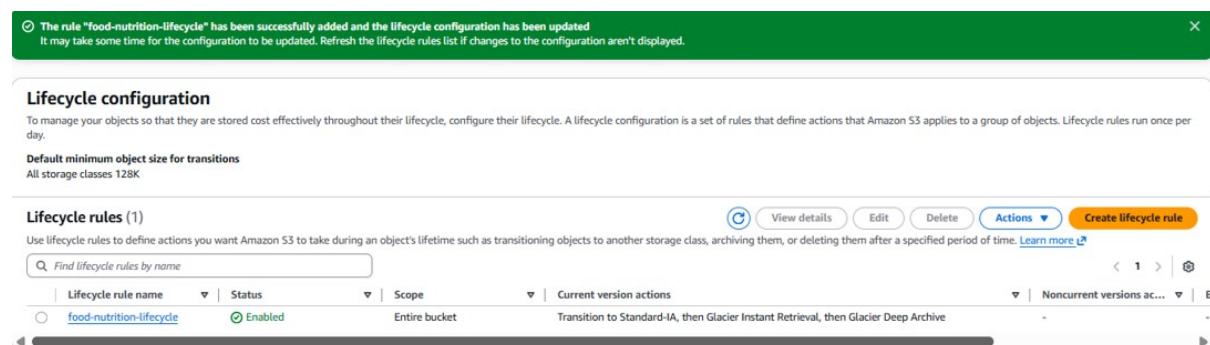


Figure 24: S3 Lifecycle Policy Configuration

6.2 Processing and Query Cost Optimization

The costs associated with Athena queries were reduced by retrieving only the necessary columns from the dataset, which lowered the amount of data being processed from 6.6 MB to 7.48 MB for each query. Although there is no considerable reduction in this small dataset, this technique saves on huge amounts when used with enterprise datasets measured in terabytes (Amazon Web Services, 2024b). Changing the file type to Parquet reduced the amount of data scanned by 87% per query.

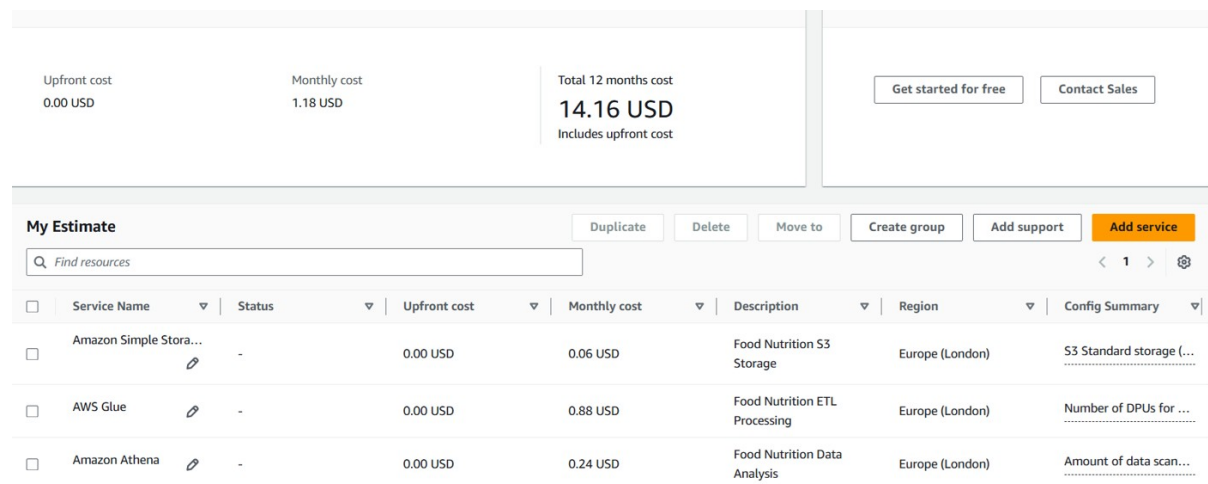


Figure 25: Athena Query Optimisation Comparison

6.3 Monitoring and Budget Management

An AWS Budget Alert has been set with a budget limit of \$10 per month with emails alerts whenever spending hits 80 percent of the budget. AWS Cost Explorer has been used for monitoring cost level at the services level such as S3, Glue, and Athena (Amazon Web Services, 2024c).

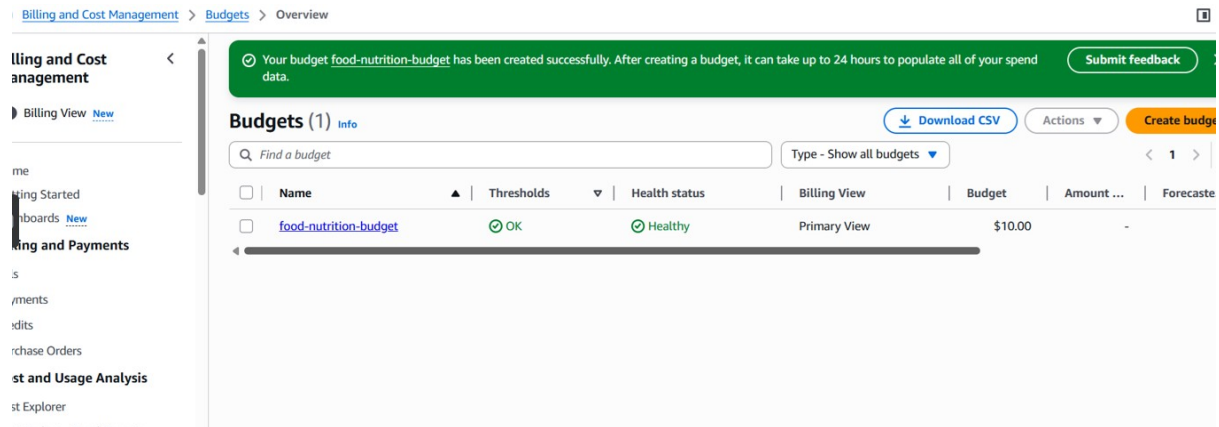


Figure 26 : AWS Budget Alert Configuration

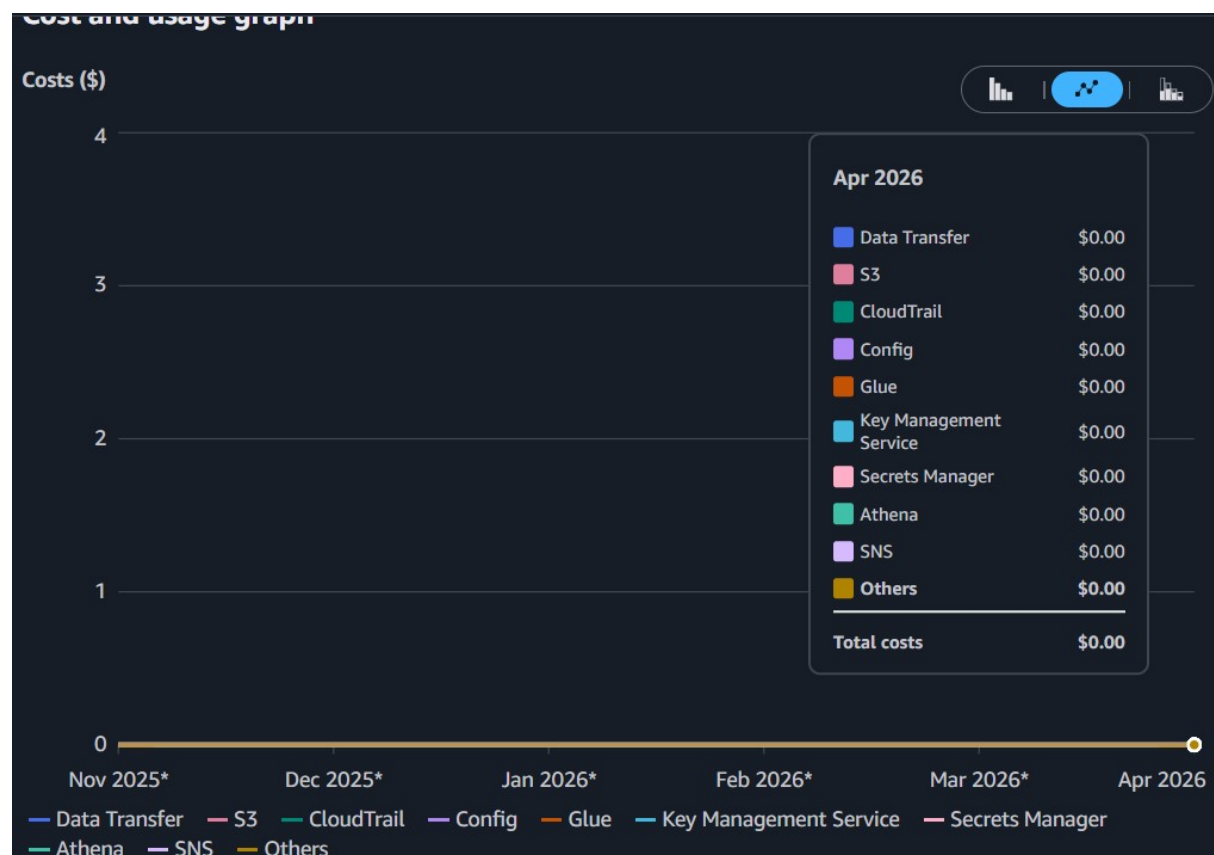


Figure 27 : AWS Cost Explorer Dashboard

7. Security and Compliance

7.1 Security Measures and Data Integrity

Strong security controls have been put in place for the cloud computing food nutrition solution, ensuring that the data confidentiality, integrity, and availability are maintained. Server-side encryption utilizing AWS-managed keys was applied to the bucket known as food-nutrition-bigdata-bdcc, ensuring that all data is encrypted when stored (Amazon Web Services, 2024a).

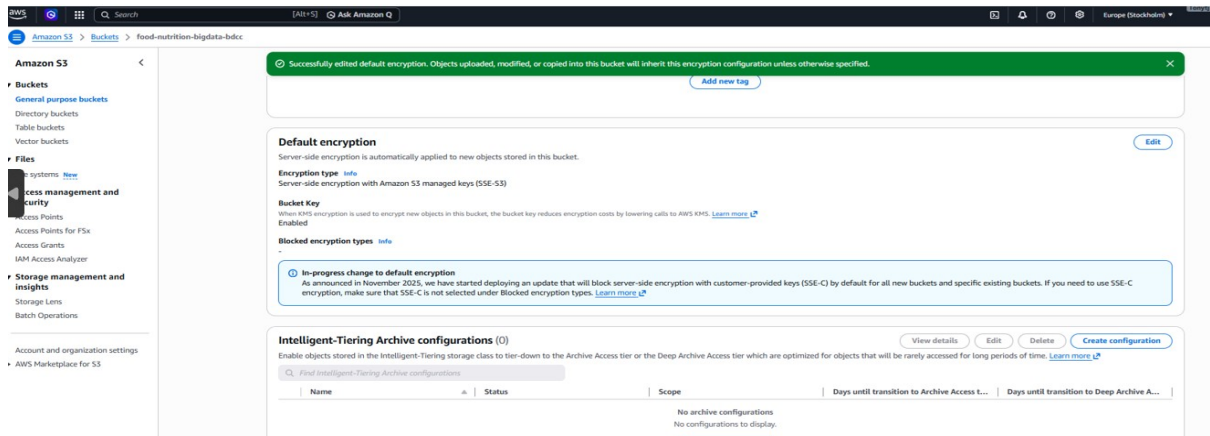


Figure 28: Amazon S3 Server-Side Encryption Configuration

All four S3 Block Public Access features were configured for the project to ensure that there could be no possible access to the food nutrition data by the public. In addition, S3 Bucket Versioning was configured for tracking and rollback purposes in case of any deletion or loss of integrity due to corruption of files (Amazon Web Services, 2024a).

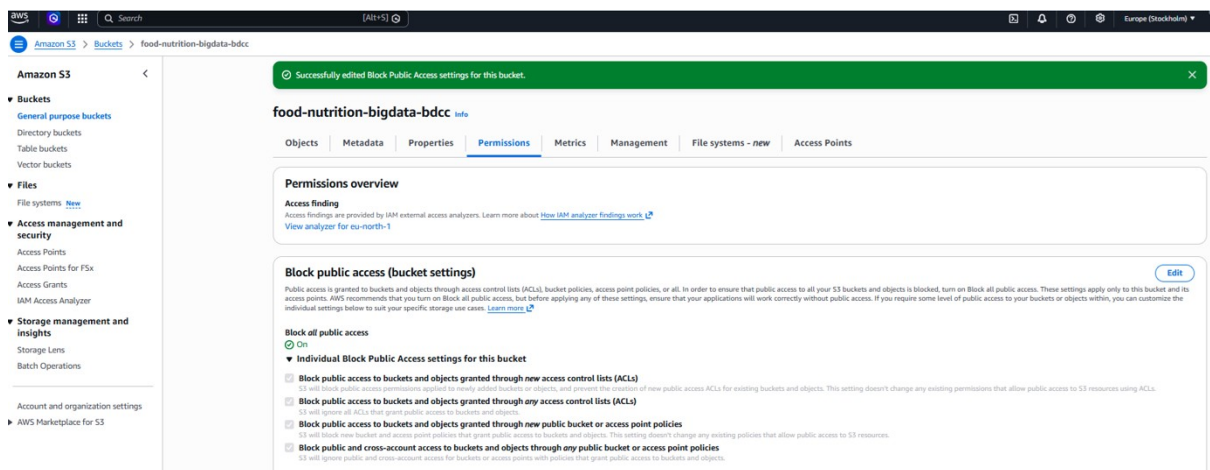


Figure 29 : S3 Block Public Access Settings

AWS CloudTrail was configured with a trail named food-nutrition-trail, recording all management events performed on the AWS account. CloudTrail logs are stored in a dedicated S3 bucket food-nutrition-cloudtrail-logs, providing a complete audit trail of all data access and configuration changes (Amazon Web Services, 2024b).

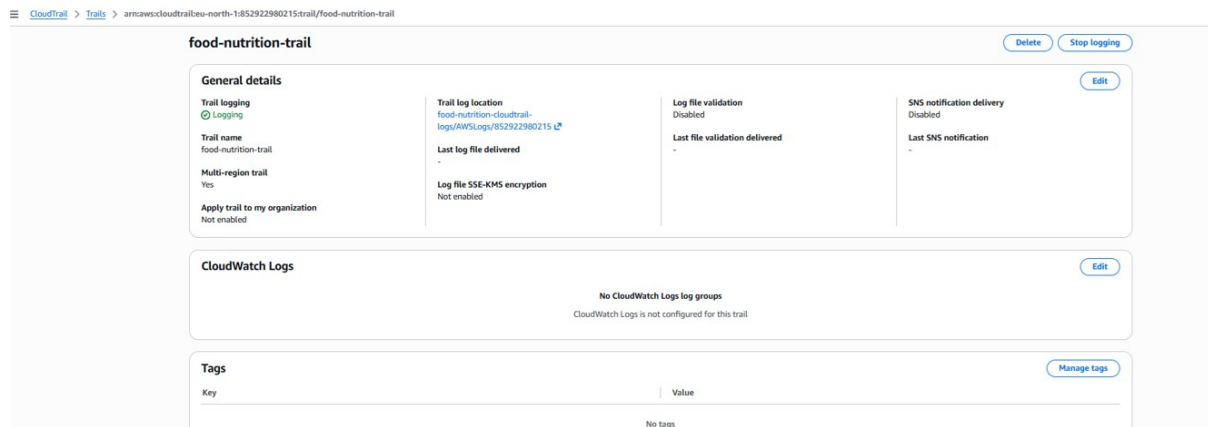


Figure 30 : AWS CloudTrail Audit Logging Configuration

7.2 Access Control and Compliance

The IAM policy was set up to ensure least privilege. An IAM user food-data-analyst was established for whom three particular policies were applied: AmazonS3ReadOnlyAccess, AmazonAthenaFullAccess, and AWSGlueConsoleFullAccess. This will ensure that the service performs only what it is supposed to do.

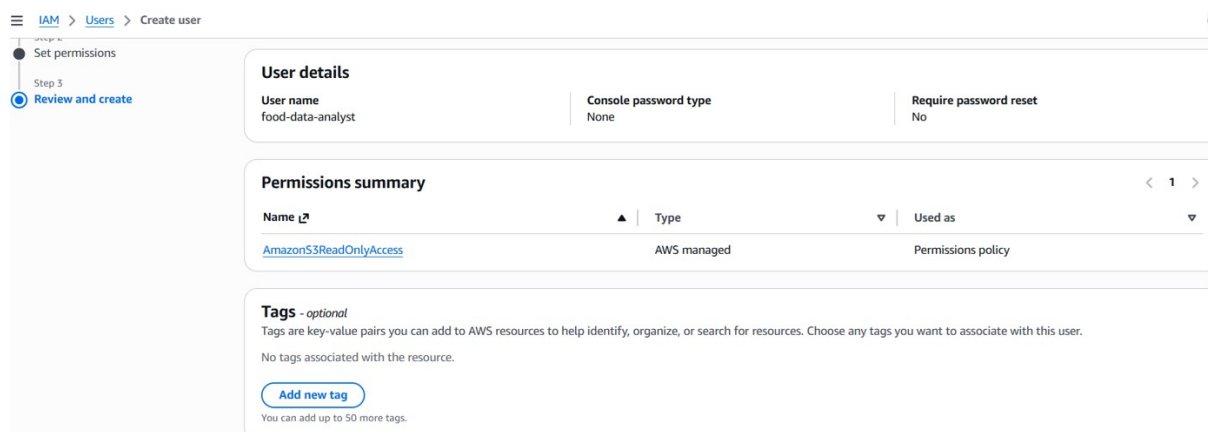


Figure 31: IAM User Access Control Policies

As far as the issues of compliance are concerned, all the data handling and storage activities were undertaken in the EU (London) eu-west-2 geographical area, thereby adhering to the demands of the UK General Data Protection Regulation (UK GDPR) pertaining to data residency and processing in authorized geographical locations (Information Commissioner's Office, 2021). All these elements form an effective security architecture consistent with enterprise-level compliance.

8. Performance Monitoring and Management

8.1 Monitoring Strategies

A monitoring plan was devised for tracking the health, performance, and utilization of resources for all AWS services involved in the food nutrition application through the Amazon CloudWatch service. A dashboard known as food-nutrition-monitoring was created on Amazon CloudWatch for viewing the performance statistics of Amazon S3, AWS Glue, and Amazon Athena (Amazon Web Services, 2024a).

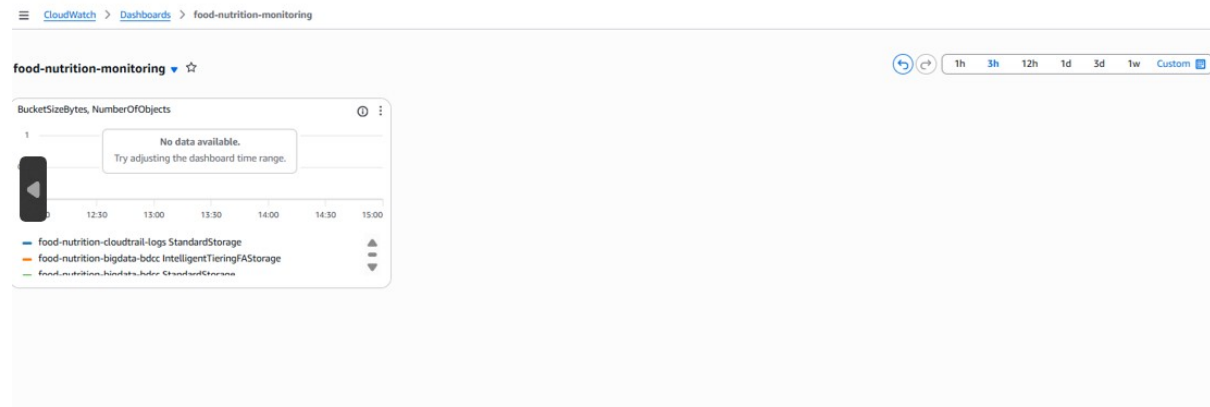


Figure 32: Amazon CloudWatch Monitoring Dashboard

An alarm was created in CloudWatch that triggered an email alert when the storage used by the S3 buckets reached 5GB, thus allowing the operations teams to manage their storage requirements beforehand without incurring any additional cost. The alarm was associated with the SNS topic called food-nutrition-alerts (Amazon Web Services, 2024a).

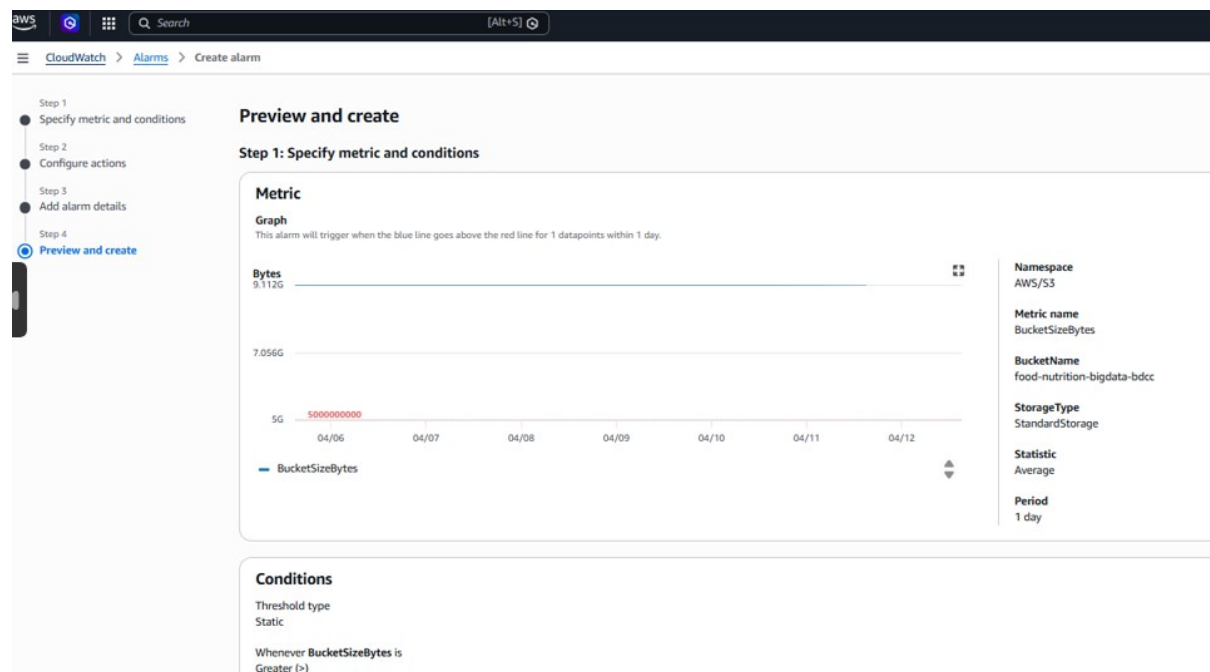


Figure 33: CloudWatch Storage Alarm Configuration

8.2 Tools and Performance Management

The AWS Glue job's efficiency was measured by monitoring the job's runs via the Glue jobs runs dashboard. The performance of the ETL job 'food-nutrition-etl-job' is successful in just 1.53 minutes with the use of 10 DPU's. There were also logs of the process which can be viewed from the Amazon CloudWatch Logs (Amazon Web Services, 2024b).

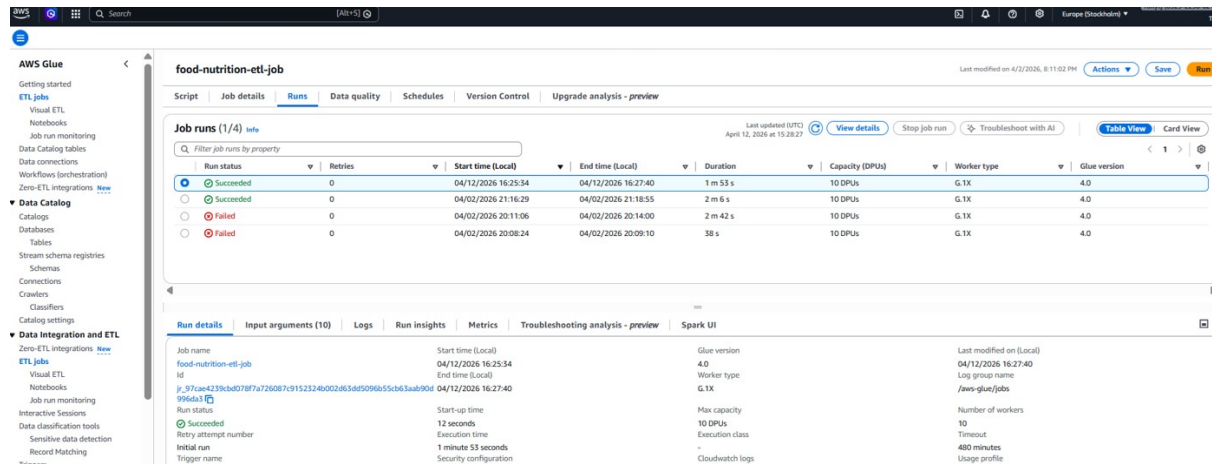


Figure 33: AWS Glue Job Performance Metrics

Query performance on Amazon Athena was checked using the CloudWatch metrics dashboard, with critical metrics being EngineExecutionTime, QueryQueueTime, and TotalExecutionTime. The performance of all queries was successful ("Succeeded"), thus demonstrating that query performance was reliable within the cleaned food nutrition dataset (Amazon Web Services, 2024a).

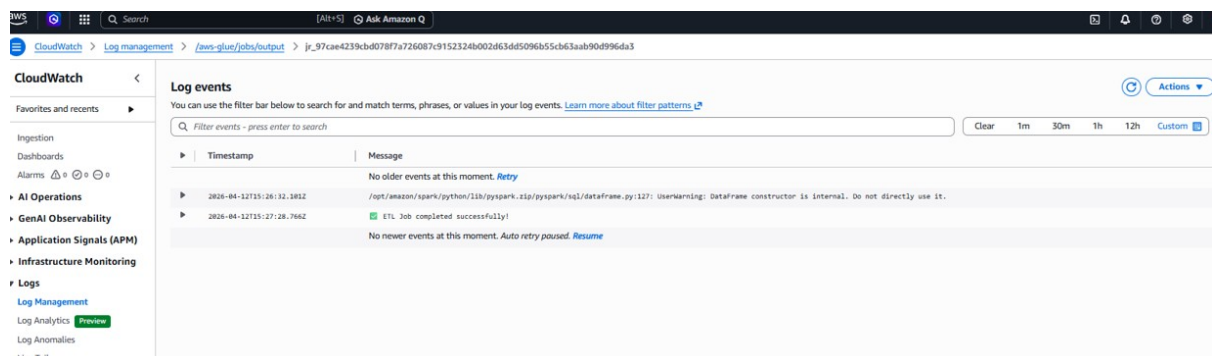


Figure 34: Amazon Athena Query Performance Metrics

AWS Config was enabled to continuously monitor and record the configuration of all AWS resources within the solution, ensuring ongoing compliance with security policies and enabling rapid identification of any configuration drift (Amazon Web Services, 2024c).

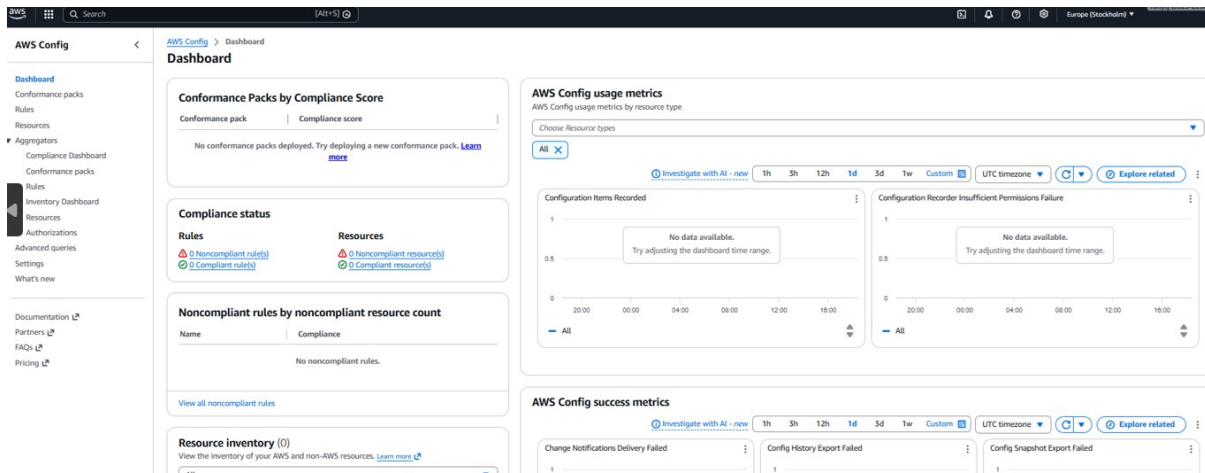


Figure 35: AWS Config Compliance Monitoring Dashboard

9. Conclusion

This report has illustrated the process of designing and implementation of an extensive big data solution to process and analyse the USDA FoodData Central database with the use of Amazon Web Services. All the aspects mentioned in the assessment brief were successfully covered by the design implemented in the project, which consisted of a fully serverless approach of Amazon S3, AWS Glue, Amazon Athena, and Amazon CloudWatch services.

It can be argued that cloud computing systems have many benefits over traditional approaches to information processing, such as scalability, affordability, and managed infrastructure (Armbrust et al., 2010). The analysis was done through five SQL queries and two machine learning algorithms and resulted in obtaining an R^2 score of 0.9663 with linear regression and finding three food category clusters with K-means.

The estimate cost of 1.18 USD per month clearly showcases the outstanding cost-effectiveness of using a serverless architecture within cloud computing for enterprise big data solutions. The security and compliance needs have been fully considered by implementing encryption, access controls, audit logging, and UK GDPR compliant data residency policies. Performance monitoring has also been implemented using CloudWatch dashboard and AWS Config tools.

In all, this case study serves to show that it is possible for big data technologies to be used via cloud computing to turn nutritional data sets into meaningful business intelligence for decision-making processes (Hashem et al., 2015).

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